

Open Source / IT — Summary (English) & MCQ Practice (QCM)

Based on the provided lecture slides (PDF). Generated on 2026-01-04.

How to use this file: Read the summary first, then try the MCQ practice without looking at the answers. Finally, check the answer key and explanations.

1. English Summary

Information Technology (IT) — Key Concepts

Definition of IT

- Information Technology (IT) is the study and use of computer systems and telecommunications to create, process, store, retrieve, and transmit information.
- IT is broader than just computers: it includes networks, software, data, and the processes used to manage information.

Data vs. Information

- **Data** are raw facts (numbers, symbols, measurements).
- **Information** is processed/organized data that has meaning and supports decision-making.

Technology and Computer Science

- **Technology** refers to tools and methods used to solve practical problems or perform tasks.
- **Computer Science** studies computation, algorithms, programming, and the theory/practice of computing systems.

Why IT Matters

- Improves efficiency and productivity by automating tasks.
- Enables global communication and collaboration.
- Drives innovation (new services, products, and business models).

IT Tools

1) Hardware

- **Hardware** is the physical part of a computer system (devices you can touch).
- A **computer** is a machine that can be programmed to automatically perform arithmetic or logical operations.

Types of Computers (examples)

- Desktop PCs (fixed computers used at home/offices).
- Laptops (portable PCs).
- Servers (powerful systems that manage and store data).
- Workstations (high-performance machines for engineers/designers).

Main Computer Components

- Motherboard, Processor (CPU), RAM, Hard Disk (HDD/SSD), Graphics Card (GPU), Network Card (NIC).

Motherboard

- Central board that connects components (CPU, RAM, storage, cards) and provides connection ports.

- Helps centralize and coordinate data exchange between components.

Processor (CPU)

- The “brain” of the computer: executes instructions and coordinates data exchange.
- Key characteristics: **cores** (parallel work), **threads** (concurrent execution), **clock speed** (frequency), **cache** (fast storage near CPU).

Memory

- **RAM**: temporary (volatile) memory used for fast access while programs run.
- **ROM**: non-volatile memory that keeps data without power; often stores startup programs (e.g., BIOS).

Storage (Hard Disk)

- Non-volatile mass storage used to store the OS, installed programs, and personal data.
- Two categories: **HDD** (magnetic, older, slower, cheaper for large capacity) and **SSD** (flash-based, faster, more reliable, more expensive).

Graphics and Networking

- **Graphics card (GPU)** handles display and complex graphics tasks (3D rendering, video, games).
- **Network card (NIC)** allows the computer to communicate on a network (Ethernet/Wi-Fi/Bluetooth).

Peripherals

- Auxiliary input/output devices that are not core components (e.g., keyboard, mouse, scanner, printer, speakers).

Communication Network Devices

- Examples: modem, hub, switch, router, bridge, gateway, repeater.
- **Switch**: connects devices in the same LAN; sends data to the right device; reduces congestion.
- **Router**: routes packets between networks (LAN → WAN); chooses paths; can provide Wi-Fi.
- **Modem**: converts digital ↔ analog signals for ISP access; links your network to the external world.

Transmission Media and Wireless

- **Fiber optics** carry data as light pulses; high-speed, reliable, good for long distances.
- **Wi-Fi** provides wireless connectivity; improves mobility and connects phones, laptops, and IoT devices.

Mobile Devices

- Small computers designed to be held/used in hand (e.g., smartphones, tablets).
- Features: portability, wireless connectivity (Wi-Fi/Bluetooth/4G/5G), touchscreens, mobile OS (Android/iOS), access to online services and apps.

Smart Chips and IoT

- Smart chips (RFID/NFC, embedded processors, IoT sensors) integrate sensing, storage, computation, and data security into low-power devices.
- Features: miniaturization, low power, wired/wireless communication, integration into objects/environments, real-time automation and traceability.

RFID (Radio Frequency Identification)

- Automatic identification using radio waves to exchange data between a tag and a reader.
- Components: RFID tag (passive/active), reader, and a database/information system.
- Advantages: fast reading, can identify multiple objects, automates processes.
- Limitations: cost, security/privacy risks, electromagnetic interference.
- Common uses: inventory/logistics, access control badges, tracking, contactless payment/ticketing.

Embedded Processors

- Computing chips integrated into larger systems to perform specific tasks.
- Applications: automotive electronics (ABS, airbags, GPS), medical devices, smart home appliances, robotics/industrial systems.

IoT Sensors

- Devices that collect physical data (temperature, motion, humidity, light, etc.) and transmit it for remote analysis/control.
- Uses: smart cities, connected agriculture, digital health, Industry 4.0.
- Advantages: intelligent automation, real-time data collection, improved efficiency.
- Limitations: dependence on connectivity, cybersecurity vulnerabilities, managing large data volumes.

2) Software

- **Software** includes system software (operating systems), application software, and office software.

Operating Systems (OS)

What an OS is

- An **operating system** is system software that controls and coordinates computer hardware and software resources and provides services for running applications.

Core OS Functions

- **Process management:** starting/stopping programs, allocating resources, managing memory usage.
- **File management:** creating, opening, closing, deleting, and organizing files on storage.
- **Security management:** access control (login), firewall, protecting memory from malicious access, vulnerability messages.
- **Memory management:** tracking how applications/processes use memory and preventing overload; freeing space when needed.

- **Device controlling:** managing connected devices (input/output/storage) and installing updates/drivers.

Program vs. Process

- A **program** is a set of instructions stored in memory (a passive, static entity).
- A **process** is a program in execution (active, dynamic, uses CPU/memory/I/O).

OS Types and Examples

- Examples: Windows, Linux, macOS.
- Operating systems can also be classified (e.g., multitasking, multi-user, real-time, embedded) depending on design goals.

Elements of an OS

- **Kernel:** essential layer loaded during boot; enables hardware/software to work together; if it fails the system may stop.
- **Hardware Abstraction Layer (HAL), libraries, and system utilities** support the OS and applications.

Kernel (more detail)

- First software loaded into memory during startup; usually protected from modification/access by other programs.
- Critical role: if the kernel stops functioning, the whole system is affected (unlike a normal application crash).

Software Applications

Business Applications

- **CRM** (Customer Relationship Management): tracks customers, manages sales processes, improves communication.
- **ERP** (Enterprise Resource Planning): integrates core operations (finance, HR, inventory, supply chain) into one system.
- **Project management tools:** plan/monitor/control projects, assign tasks, track progress, and support team communication.

Office Software (Office Suites)

- Office software helps create/edit/share documents, spreadsheets, and presentations, often with collaboration features.
- Well-known example: Microsoft Office (Word, Excel, PowerPoint, Outlook, OneNote, Access).
- Trends: cloud services and subscription models (e.g., Microsoft 365) with continuous updates.

Pros and Cons (Example: Microsoft Office)

- **Pros:** familiarity/compatibility, advanced features, cloud integration, updates & support, data security, cross-platform.
- **Cons:** can be costly, large installation size, limited free alternatives for full functionality, cloud features depend on Internet.

Applications of IT

Communication Spaces

- Three types: **Internet** (global public connectivity), **Intranet** (restricted within an organization), **Extranet** (controlled access for external partners/clients).
- Used to create virtual/physical environments for interaction, collaboration, and information exchange (video conferencing, collaborative tools, messaging, social media, virtual workspaces).
- Supports telemedicine, e-learning, e-government, e-commerce, and remote work.

Multimedia

- Computer-based applications integrating text, graphics, images, audio, animation, and video.
- Used for communication, education, entertainment, business, medicine, advertising, conferencing, and science.

Electronic Data Interchange (EDI)

- Computer-to-computer exchange of standardized business documents between organizations.
- Automates processes like order processing, invoicing, inventory management, shipping, and payment remittances.
- Benefits: efficiency, cost savings, faster transactions, improved accuracy, better supply chain visibility, stronger partner relationships, scalability.

Workflows

- Use of specialized software to automate, manage, and optimize business/IT processes.
- Key applications: repetitive task automation, IT service management, process standardization, cross-department collaboration, project/task management, process monitoring & reporting.

2. MCQ Practice (QCM)

Choose the **one best** answer for each question.

Q1. Which statement best describes Information Technology (IT)?

- **A.** The study of mathematical proofs only
- **B.** The study and use of computer systems and telecommunications to create, process, store, retrieve, and transmit information
- **C.** The physical parts of a computer only
- **D.** A programming language used for web development

Q2. In the course context, what is the best definition of **data**?

- **A.** Processed facts that already have meaning
- **B.** Raw facts such as numbers, symbols, or measurements
- **C.** A set of instructions stored in memory
- **D.** A device that routes packets between networks

Q3. Which option is an example of **information** rather than data?

- **A.** A list of temperatures: 20, 21, 19, 22
- **B.** A database table of raw sensor readings
- **C.** A report summarizing temperature trends and highlighting unusual values
- **D.** A single number: 1500

Q4. Which of the following is **NOT** a main computer component listed in the slides?

- **A.** Motherboard
- **B.** Random Access Memory (RAM)
- **C.** Graphics card
- **D.** Optical microscope

Q5. What is the main role of the motherboard?

- **A.** To permanently store user files
- **B.** To centralize connections and coordinate data exchange between hardware components
- **C.** To convert digital signals to analog signals
- **D.** To provide wireless Internet access by itself

Q6. Which statement about the CPU is correct?

- **A.** It is a temporary (volatile) storage component
- **B.** It is the brain of the computer and executes program instructions
- **C.** It stores startup programs such as BIOS
- **D.** It is a device used only for printing

Q7. Which CPU characteristic refers to parallel processing units inside the processor that improve multitasking?

- **A.** Cache
- **B.** Clock speed
- **C.** Cores
- **D.** Hard disk

Q8. Which memory type is **volatile** (temporary) and allows quick access while programs run?

- **A.** ROM
- **B.** RAM
- **C.** HDD
- **D.** SSD

Q9. What is a key purpose of ROM in the slides?

- **A.** To store large videos permanently
- **B.** To keep data even without power, such as startup programs (e.g., BIOS)
- **C.** To render 3D graphics
- **D.** To connect to Wi-Fi

Q10. Which statement correctly compares HDD and SSD?

- **A.** HDD is flash-based and faster; SSD uses spinning platters and is slower
- **B.** HDD is magnetic with spinning platters; SSD uses flash memory and is generally faster and more reliable
- **C.** Both are volatile memories used only for temporary storage
- **D.** SSD is always cheaper than HDD for large capacity

Q11. What is the main function of a graphics card (GPU)?

- **A.** Manage file creation and deletion
- **B.** Allow the computer to display images and handle complex graphics tasks
- **C.** Convert digital data to analog over phone lines
- **D.** Direct packets between networks

Q12. What does a network card (NIC) do?

- **A.** Stores the operating system
- **B.** Allows the computer to communicate with other devices on a network
- **C.** Replaces the CPU in multitasking
- **D.** Creates spreadsheet documents

Q13. A peripheral device is best described as:

- **A.** A core component like the CPU or motherboard
- **B.** An auxiliary device used to transfer information externally (often input/output) and not a core component
- **C.** A type of operating system
- **D.** A type of fiber optic cable

Q14. Which network device connects multiple devices within the same LAN and sends data only to the intended device?

- **A.** Switch
- **B.** Modem
- **C.** Router
- **D.** Hard disk

Q15. Which network device routes data packets between different networks (LAN → WAN) and selects the best path?

- **A.** Router
- **B.** Printer
- **C.** RAM
- **D.** Scanner

Q16. What is the main role of a modem according to the slides?

- **A.** Convert digital data into analog signals (and vice versa) for transmission over telephone/cable lines
- **B.** Store application files
- **C.** Provide a graphical user interface
- **D.** Render multimedia graphics

Q17. Fiber optic cables transmit data primarily as:

- A. Electrical signals through copper
- B. Pulses of light through glass or plastic fibers
- C. Magnetic fields inside hard disks
- D. Sound waves through air

Q18. Wi-Fi is best described as:

- A. A wired networking technology that requires physical cables
- B. A wireless networking technology that connects devices without physical cables
- C. A type of permanent storage device
- D. A kind of ROM used at boot

Q19. Which is a typical feature of mobile devices in the slides?

- A. They cannot connect to the Internet
- B. They are always larger than desktops
- C. They support wireless connectivity (Wi-Fi/Bluetooth/4G/5G) and run mobile operating systems
- D. They can only be used with a wired keyboard

Q20. Smart chips in the slides include all of the following **except**:

- A. RFID / NFC
- B. Embedded processors
- C. IoT sensors
- D. Mechanical gears

Q21. RFID stands for:

- A. Rapid File Interface Design
- B. Radio Frequency Identification
- C. Real-time Fiber Internet Distribution
- D. Remote Function In Device

Q22. Which set contains the main components of an RFID system?

- A. RFID tag, RFID reader, database/information system
- B. CPU, RAM, ROM, hard disk
- C. Router, switch, modem
- D. Keyboard, mouse, printer

Q23. Which is **one limitation** of RFID listed in the slides?

- A. It cannot identify multiple objects
- B. Security and privacy risks
- C. It requires no infrastructure
- D. It never experiences interference

Q24. Embedded processors are best defined as:

- A. General-purpose laptops used for gaming
- B. Chips integrated into larger systems to perform specific tasks
- C. Optical cables used in networking
- D. A type of office software

Q25. IoT sensors mainly collect and transmit:

- A. Only text documents
- B. Physical data such as temperature, motion, humidity, or light
- C. Video games
- D. Operating system kernels

Q26. Which of the following is a common **advantage** of IoT sensors mentioned?

- A. No need for any network connectivity
- B. Intelligent automation and real-time data collection
- C. They eliminate cybersecurity risks
- D. They reduce data volume to zero

Q27. Software components in the slides include:

- A. Only hardware devices
- B. Only office furniture
- C. Operating systems, application software, and office software
- D. Only modems and routers

Q28. Which statement best defines an operating system (OS)?

- A. A device used to connect computers
- B. System software that controls and coordinates hardware/software resources and provides services for applications
- C. A document format used in offices
- D. A physical chip on the motherboard

Q29. Which is **NOT** listed as an OS function in the slides?

- A. Process management
- B. File management
- C. Security management
- D. Cooking management

Q30. A **process** is:

- A. A program in execution (active entity)
- B. A set of instructions stored on disk only
- C. A physical device
- D. A type of hard disk

Q31. A **program** is best described as:

- A. A dynamic entity created during execution
- B. A static set of instructions written to perform a task and stored in memory
- C. A router connected to a LAN
- D. A kernel crash

Q32. Which of the following is an example of OS security management in the slides?

- A. Playing multimedia videos
- B. Keeping the firewall active and requiring login
- C. Changing screen resolution only
- D. Creating spreadsheets

Q33. What is the kernel in an operating system?

- A. A word-processing document
- B. The essential OS layer loaded during boot that enables hardware/software to work together
- C. A type of peripheral device
- D. A network cable

Q34. Which statement about the kernel is correct based on the slides?

- A. If it stops functioning, the computer can continue normally
- B. It is protected from access/modification by other programs and is critical to system operation
- C. It is stored only in the keyboard
- D. It is the same as an office suite

Q35. CRM stands for and is used mainly to:

- **A.** Customer Relationship Management; track customers and manage sales/communication
- **B.** Computer Resource Management; manage CPU cores
- **C.** Customer Router Module; connect LAN to WAN
- **D.** Cloud RAM Model; increase memory speed

Q36. EDI is best defined as:

- **A.** A multimedia application for entertainment
- **B.** Computer-to-computer exchange of standardized business documents between organizations
- **C.** A type of ROM used at boot
- **D.** A wireless network protocol

Q37. Which is a benefit of EDI mentioned in the slides?

- **A.** More manual data entry
- **B.** Reduced efficiency
- **C.** Improved data accuracy through standardized formats
- **D.** Slower transactions

Q38. Which communication space provides restricted access within an organization?

- **A.** Internet
- **B.** Intranet
- **C.** Extranet
- **D.** Bluetooth

Q39. Workflows in IT refer to:

- **A.** Only gaming applications
- **B.** Specialized systems to automate, manage, and optimize business and IT processes
- **C.** A type of storage device
- **D.** A programming language

3. Answer Key & Explanations

Q1 — Correct answer: B

IT covers systems and technologies used to handle information (computers + telecommunications + processes).

Q2 — Correct answer: B

Data are raw facts; information is processed/organized data with meaning.

Q3 — Correct answer: C

Information is processed/organized data that supports understanding or decisions.

Q4 — Correct answer: D

Motherboard, CPU, RAM, hard disk, graphics card, and network card are main components; a microscope is not.

Q5 — Correct answer: B

The motherboard is the main board that connects components and supports their communication.

Q6 — Correct answer: B

The CPU executes instructions and coordinates operations; RAM is temporary, ROM stores BIOS.

Q7 — Correct answer: C

Cores act as processing units inside the CPU and allow more tasks at once.

Q8 — Correct answer: B

RAM is temporary/volatile; ROM and disks are non-volatile.

Q9 — Correct answer: B

ROM is non-volatile and commonly stores startup code like BIOS.

Q10 — Correct answer: B

The slides describe HDD as older magnetic and SSD as newer flash-based, faster and more reliable but more expensive.

Q11 — Correct answer: B

GPU supports display and heavy graphics workloads such as rendering and video.

Q12 — Correct answer: B

A NIC enables network communication; it may connect via Ethernet (RJ45) or wireless variants.

Q13 — Correct answer: B

Peripherals are external/auxiliary hardware controlled by the computer (keyboard, mouse, printer, etc.).

Q14 — Correct answer: A

A switch connects devices in a LAN and improves efficiency by forwarding data to the right destination.

Q15 — Correct answer: A

A router connects networks and directs packets across them.

Q16 — Correct answer: A

A modem performs modulation/demodulation and provides ISP access.

Q17 — Correct answer: B

Fiber optics carry data as light pulses and offer high bandwidth and reliability.

Q18 — Correct answer: B

WiFi enables wireless connectivity and mobility.

Q19 — Correct answer: C

Mobile devices are portable and support wireless connectivity and mobile OSs.

Q20 — Correct answer: D

Smart chips are electronic components like RFID, embedded processors, and IoT sensors.

Q21 — Correct answer: B

RFID is Radio Frequency Identification.

Q22 — Correct answer: A

The slides list tag, reader, and database/information system.

Q23 — Correct answer: B

Limitations include cost, security/privacy risks, and electromagnetic interference.

Q24 — Correct answer: B

Embedded processors are integrated into machines/devices (cars, appliances, medical equipment).

Q25 — Correct answer: B

IoT sensors collect physical/environmental measurements and send them for analysis/control.

Q26 — Correct answer: B

Advantages include intelligent automation, real-time data collection, and improved efficiency.

Q27 — Correct answer: C

The slides state software includes OS, application software, and office software.

Q28 — Correct answer: B

An OS manages resources and provides essential services for running applications.

Q29 — Correct answer: D

The OS functions listed include process, file, communication, security, memory, and device controlling.

Q30 — Correct answer: A

The slides define a process as a program in execution.

Q31 — Correct answer: B

A program is a passive/static set of instructions; it becomes a process when executing.

Q32 — Correct answer: B

Security management includes login access control and firewall protection.

Q33 — Correct answer: B

The kernel is loaded at startup and is critical for system operation.

Q34 — Correct answer: B

The slides emphasize the kernel's critical role and protection in memory.

Q35 — Correct answer: A

CRM supports customer tracking and sales/communication processes.

Q36 — Correct answer: B

EDI automates standardized document exchange (orders, invoices, etc.).

Q37 — Correct answer: C

EDI improves accuracy, efficiency, speed, and visibility and can reduce costs.

Q38 — Correct answer: B

An intranet is internal and restricted to organization members.

Q39 — Correct answer: B

Workflow systems support automation, standardization, monitoring, and collaboration.